CHAPTER III

Plate Tectonics Theory

Prof. Dr. Rafat Zaki

Continental drift theory

نظرية الانجراف القارى

-In 1929, Alfred Wegener “The Origin of the Continents and Oceans”

-initial large super continental landmass called Pangea.

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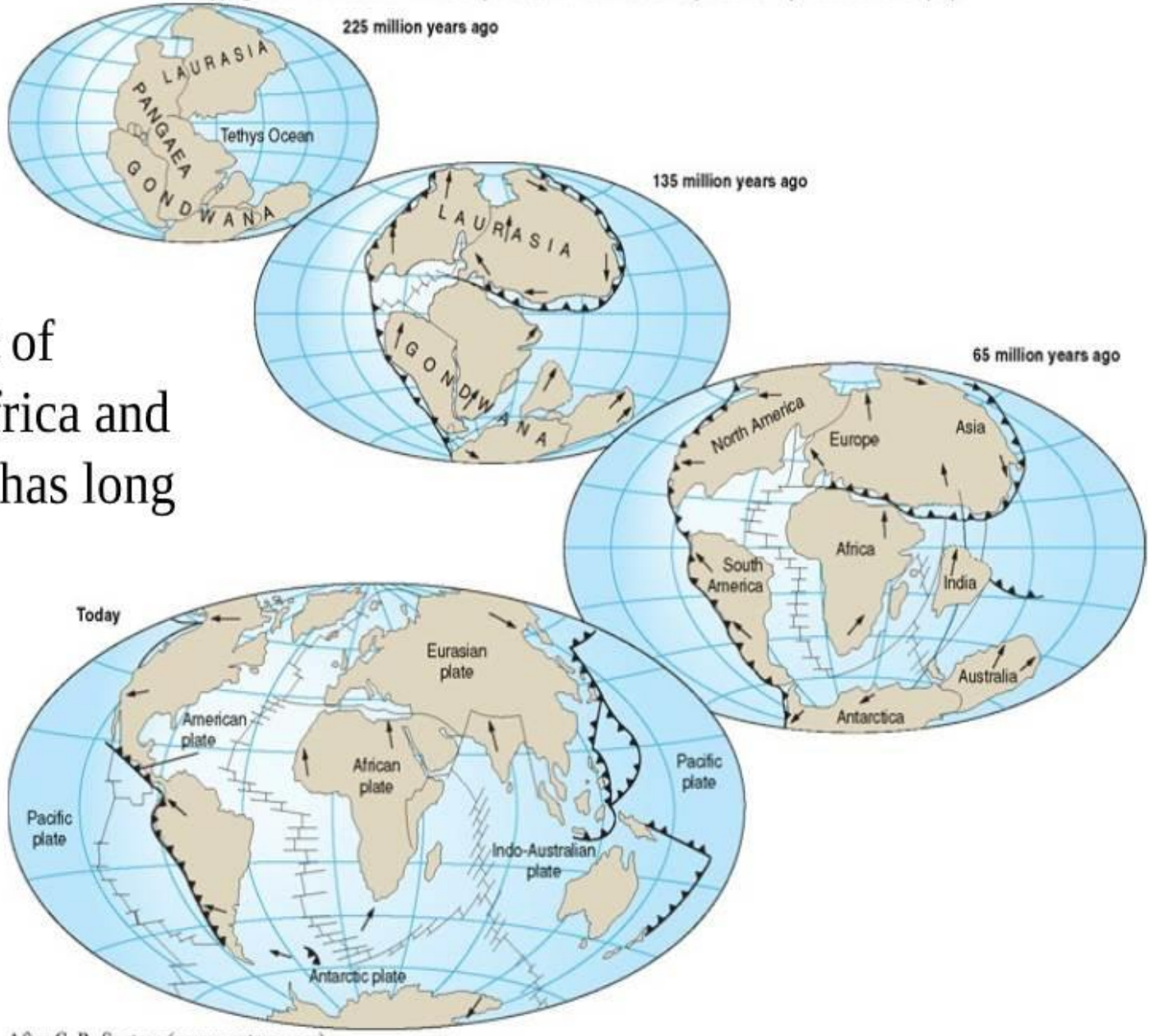
Puzzle-piece fit of coastlines of Africa and South America has long been known

Pangaea
Gondwana
Lurasia

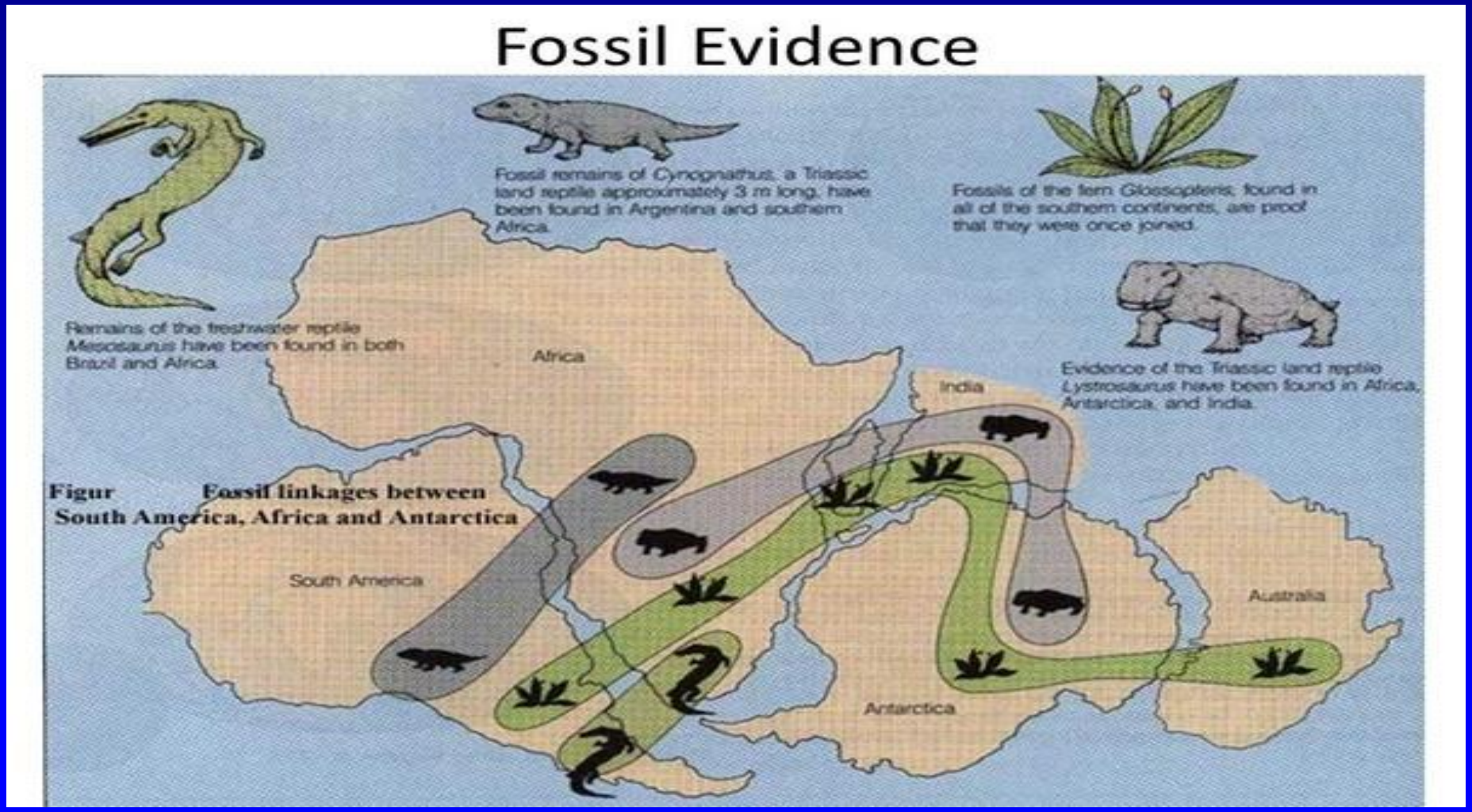


After C. R. Scotese (www.scotese.com)

Early evidences of Continental drift

1.Paleontological Evidence :شواهد حفرية قديمة

Glossopteris: is a fossil plant



3. Glacial Evidence شواهد طقس ثلجي

-During the Paleozoic Era (*300 Ma), The continents of the southern hemisphere show evidence of glaciation.

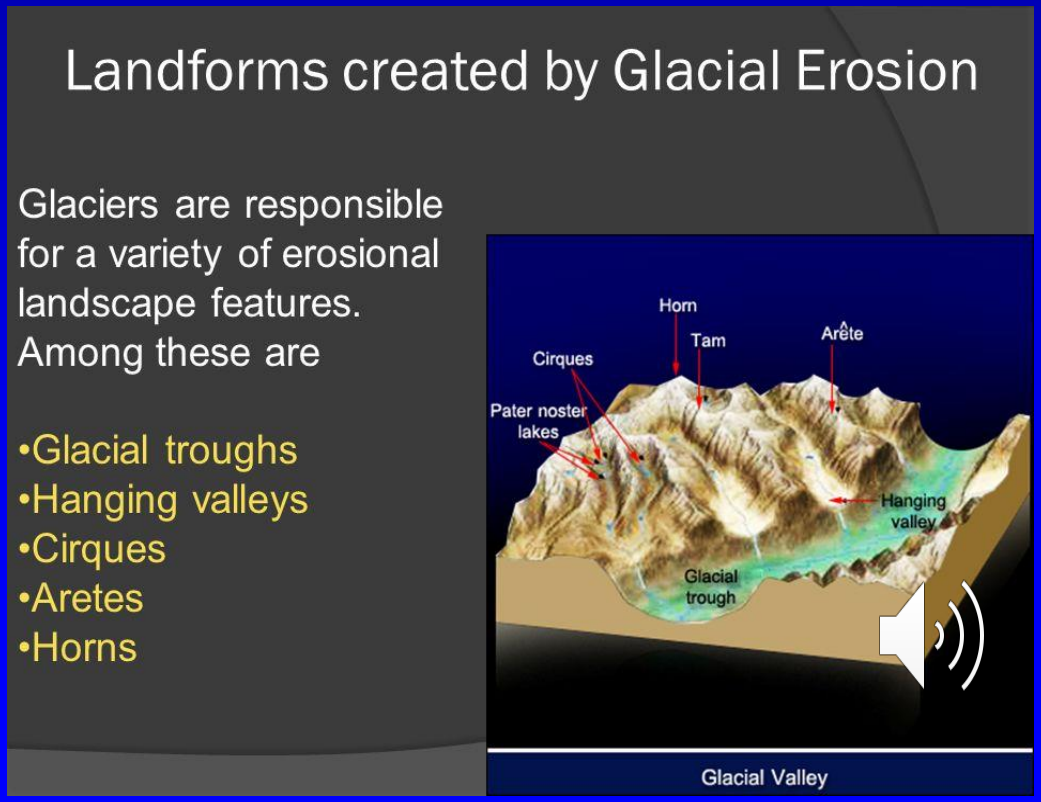
- The striations (علامات الخدوش) that glaciers leave as they scrape across bedrock were confusing.
- these straiations and glacial deposits are consistent with a Pangaean glaciations at the South pole.

3- Paleoclimatic Evidence

شواهد مناخية قديمة

- Evaporites, sandstone, coral reefs and coal

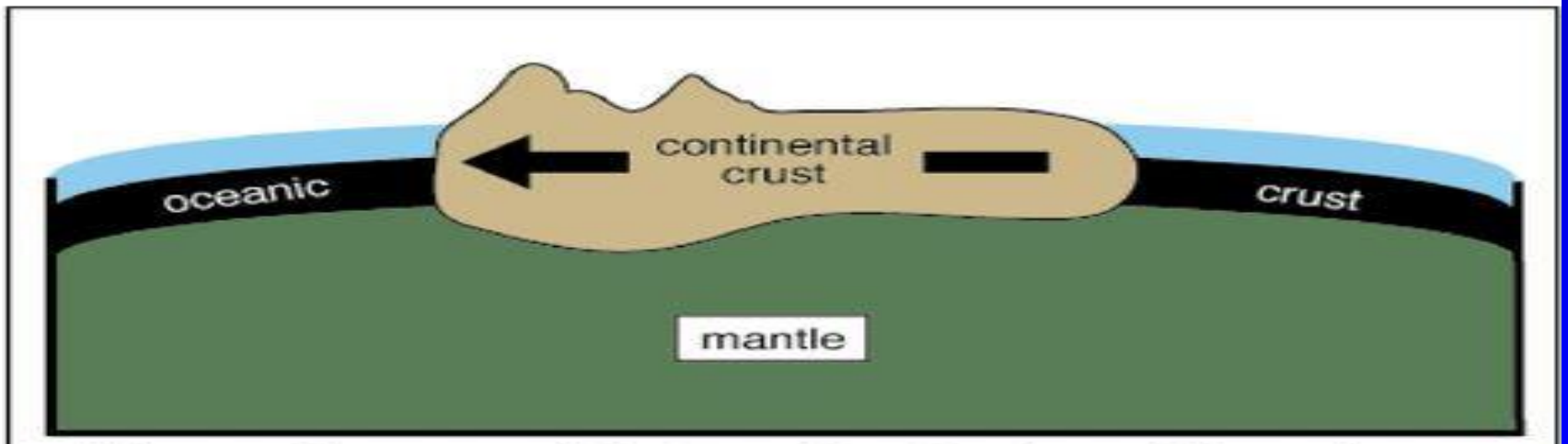
التشابه المناخي بين منطقتين متعاكستي المناخ تمامًا، وقد تمّ إثبات ذلك باكتشاف بعض الرواسب من الشعب المرجانية أو الفحم المرتبط تواجدها بالمناطق الحارة قديمًا، ولكن محل تواجدها الحالي في المناطق الباردة كشمال كندا وسيبيريا



Coal beds of North America and Europe indicate Laurasia super continent.

Continental Drift Problems

- Wegener could not come up with viable *driving force.*
- Continents should not be able to "plow through" sea floor rocks.

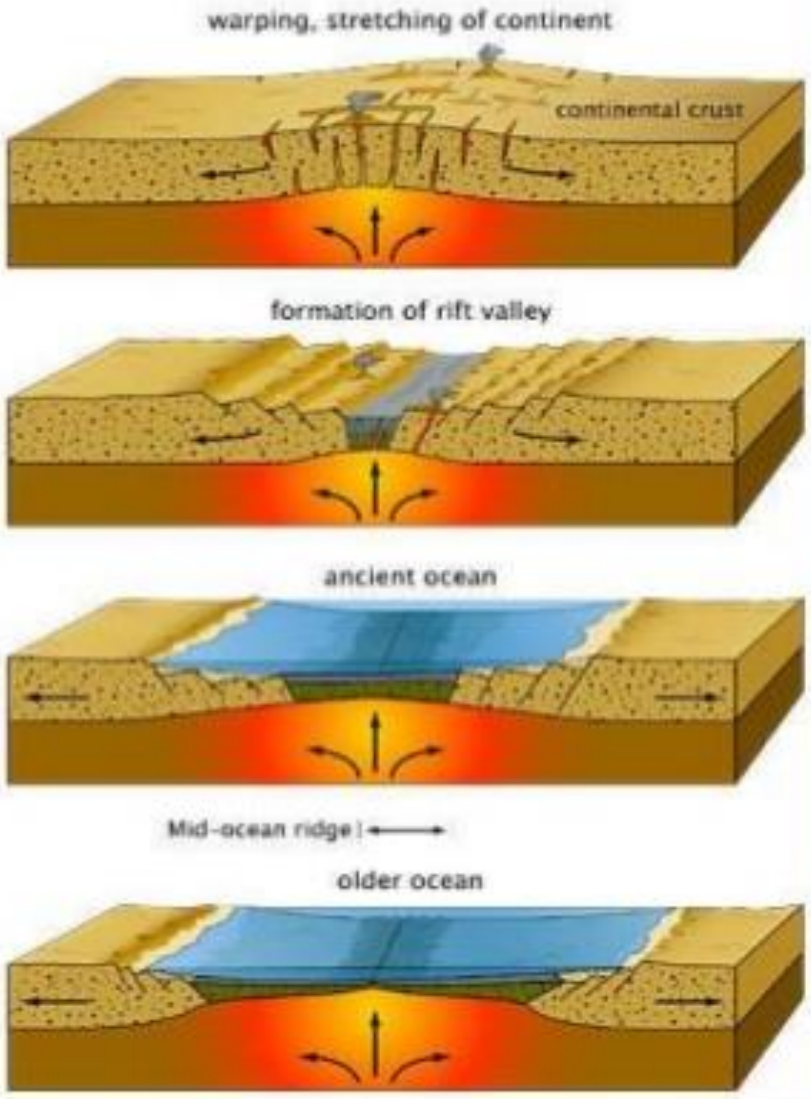


Wegener's proposal that continents plowed through oceanic crust was not accepted by other geologists.

Seafloor spreading (Hess. 1962)

نظرية انتشار قاع المحيط

- The Earth's **lithospheric plates** move along with **convection currents** in the mantle **rifting** plates apart along the mid-ocean ridge.
- Molten material **erupts** through the valley that runs along the center of the mid-ocean ridge.
- As **hot magma** is forced **upward**, the sea floor **spreads apart**.
- This material eventually **cools**, **sinks**, and **hardens** to form the high ridged rock of the ocean floor.

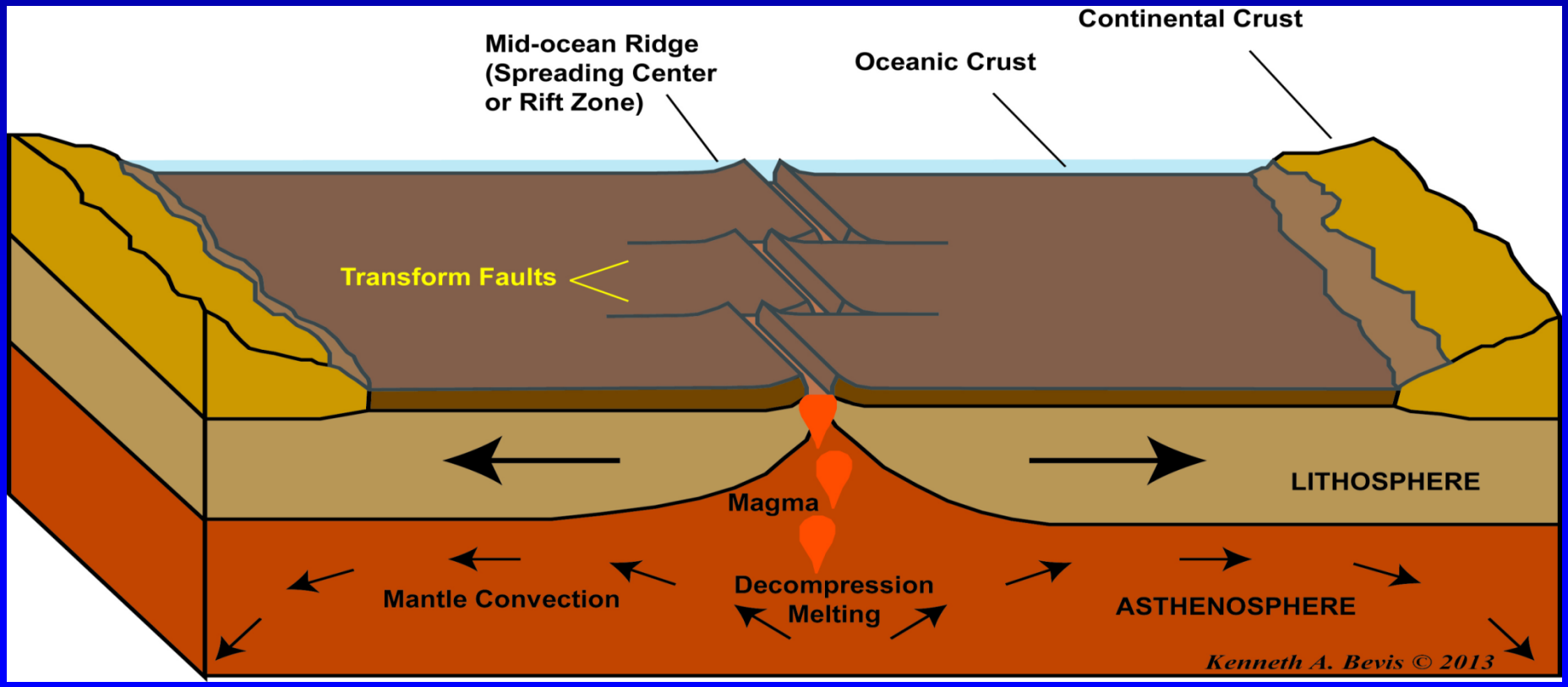


Types of Plate Motion (Wilson, 1965)

انواع حركات الصفائح

1- Divergent Plate Boundary حدود تباعدية

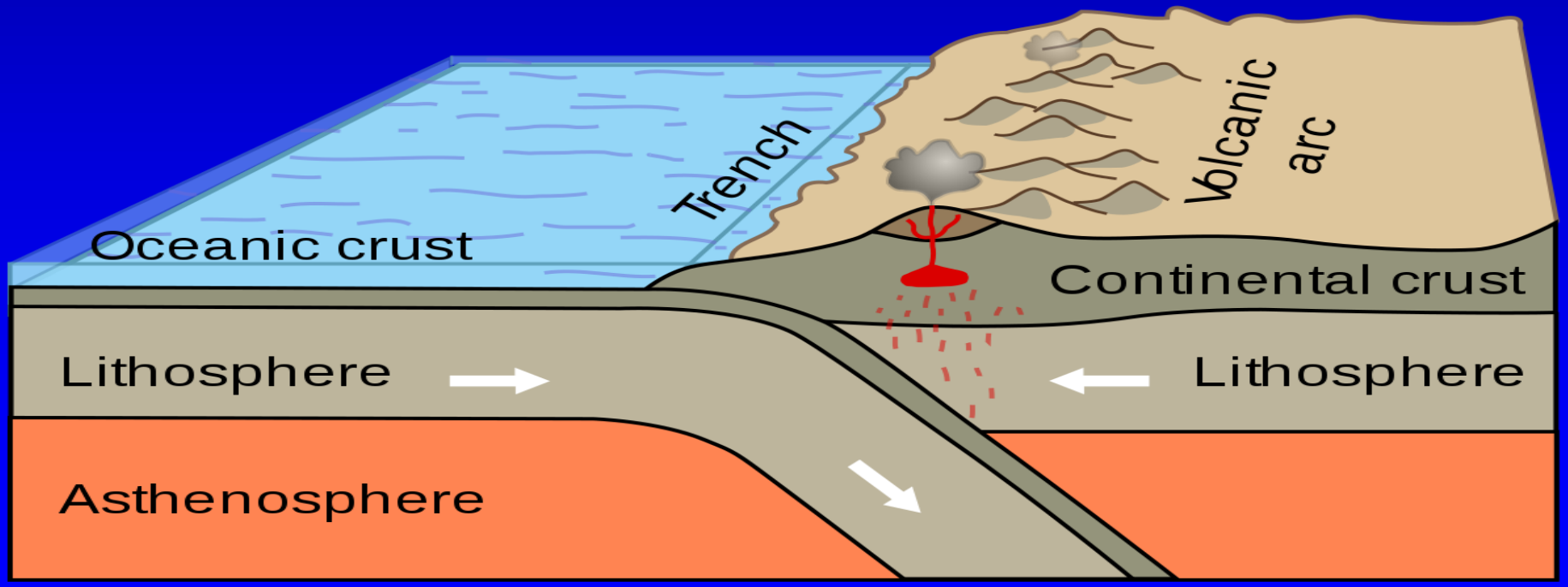
where plates move away from each other.



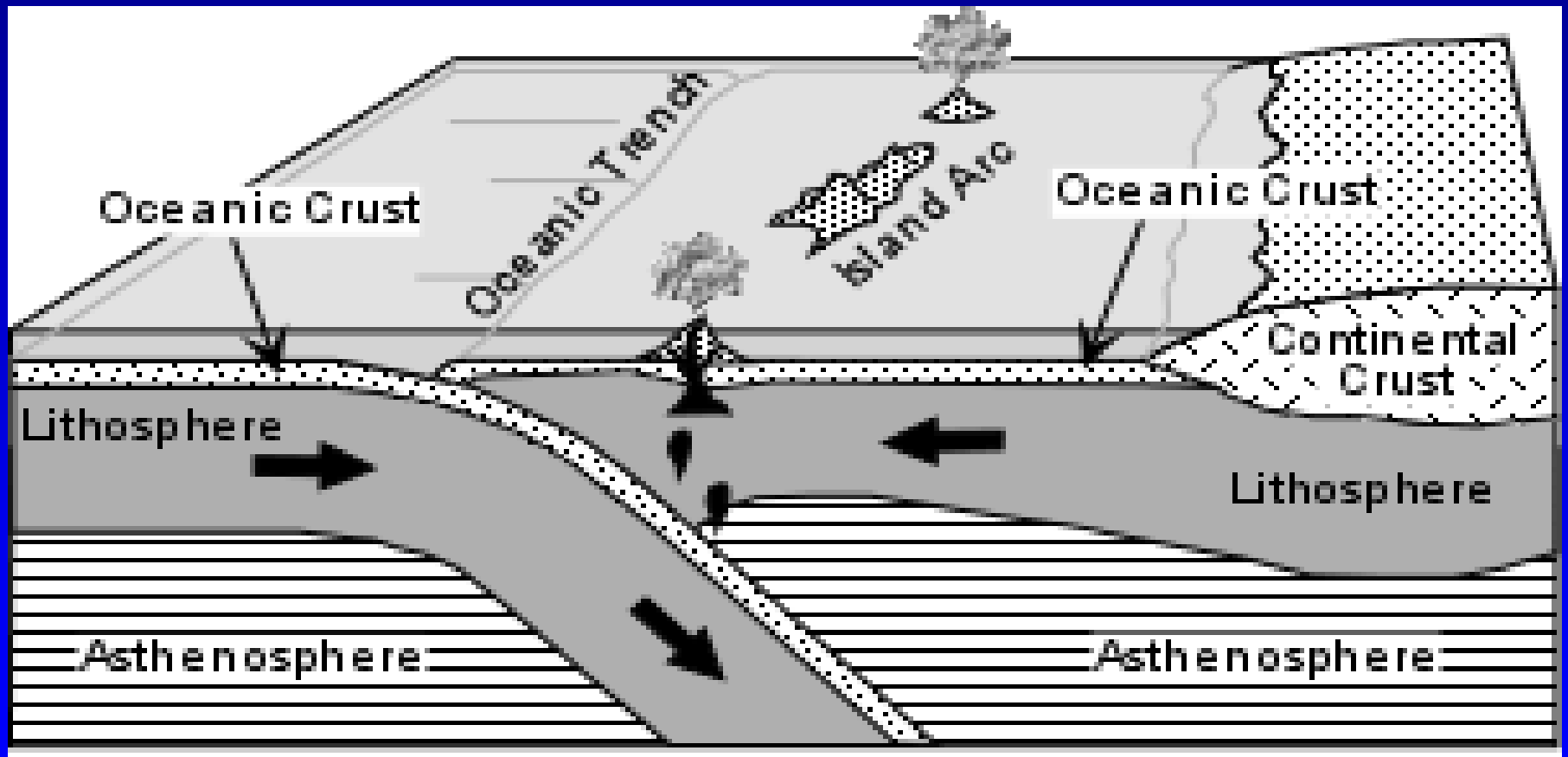
Bur Sa'id
Mansura
Mahalla el Kubra
Zagazig
Ismailiya
Al Qahirah
As Suways
EGYPT
SINAI
Israel
Jordan
SAUDI ARABIA
Asyut
Sohag
Qena
Al Uqsur

Convergent Plate Boundaries حدود تقاربية

- Plates move toward each other
- Mountain belts and volcanoes common
- Oceanic plates may sink along a ***subduction zone***, الاندساس
- Typically marked by a deep ocean trench خندق



- *Ocean-ocean plate convergence* (Marked by *ocean trench*,, and *volcanic island arc*)
- تقارب صفيحة المحيط والمحيط تتميز بخندق المحيط وقوس الجزيرة البركانية



Ocean - Ocean Convergence

- *Ocean-continent plate convergence* (Marked by ocean trench, *volcanic arc*, and *mountain belt*
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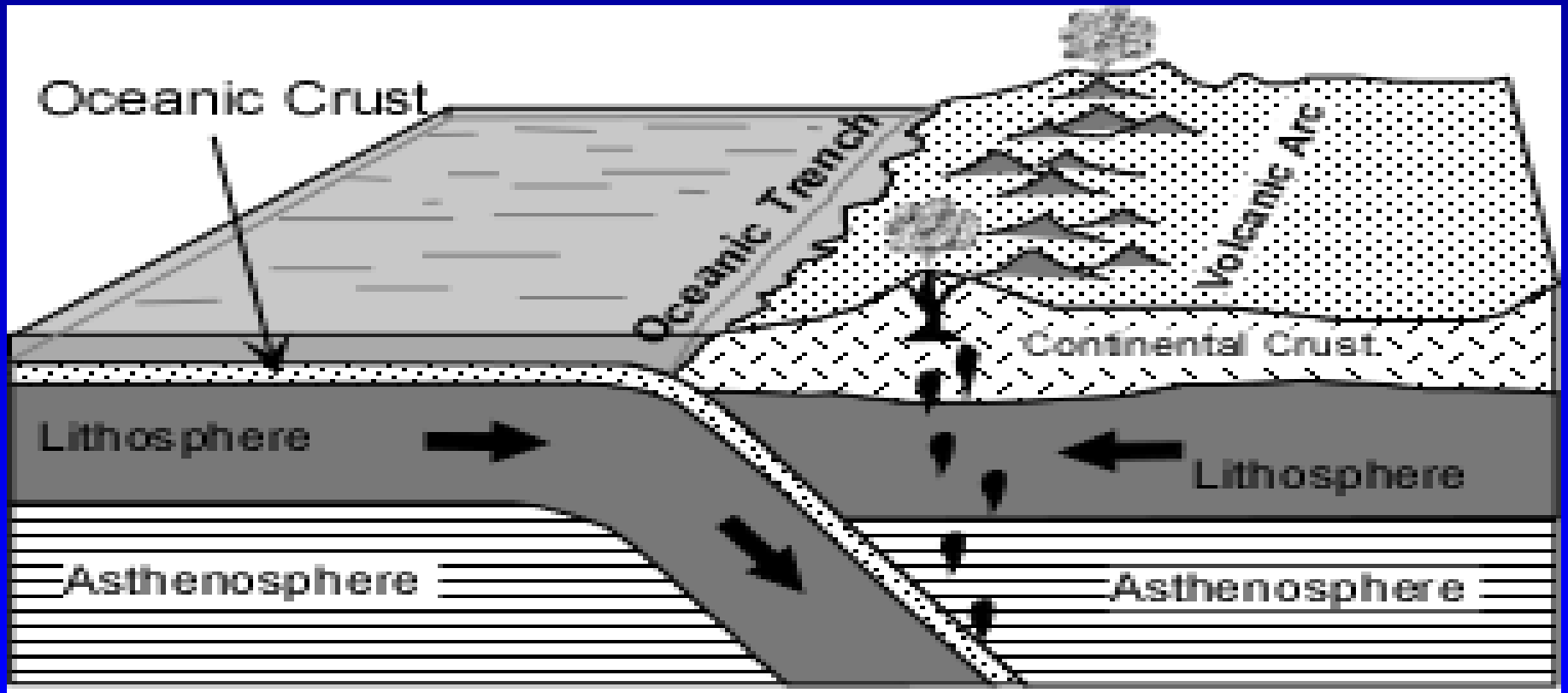
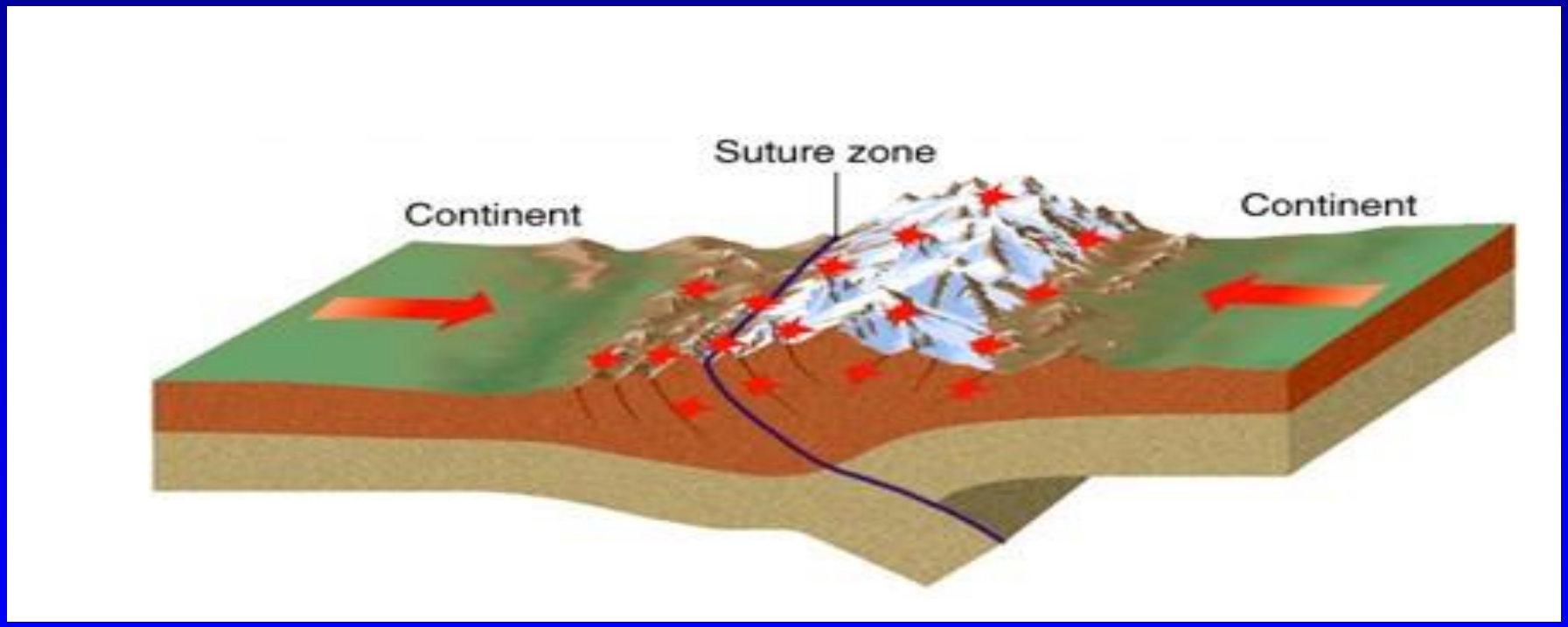


Ocean - Continent Convergence

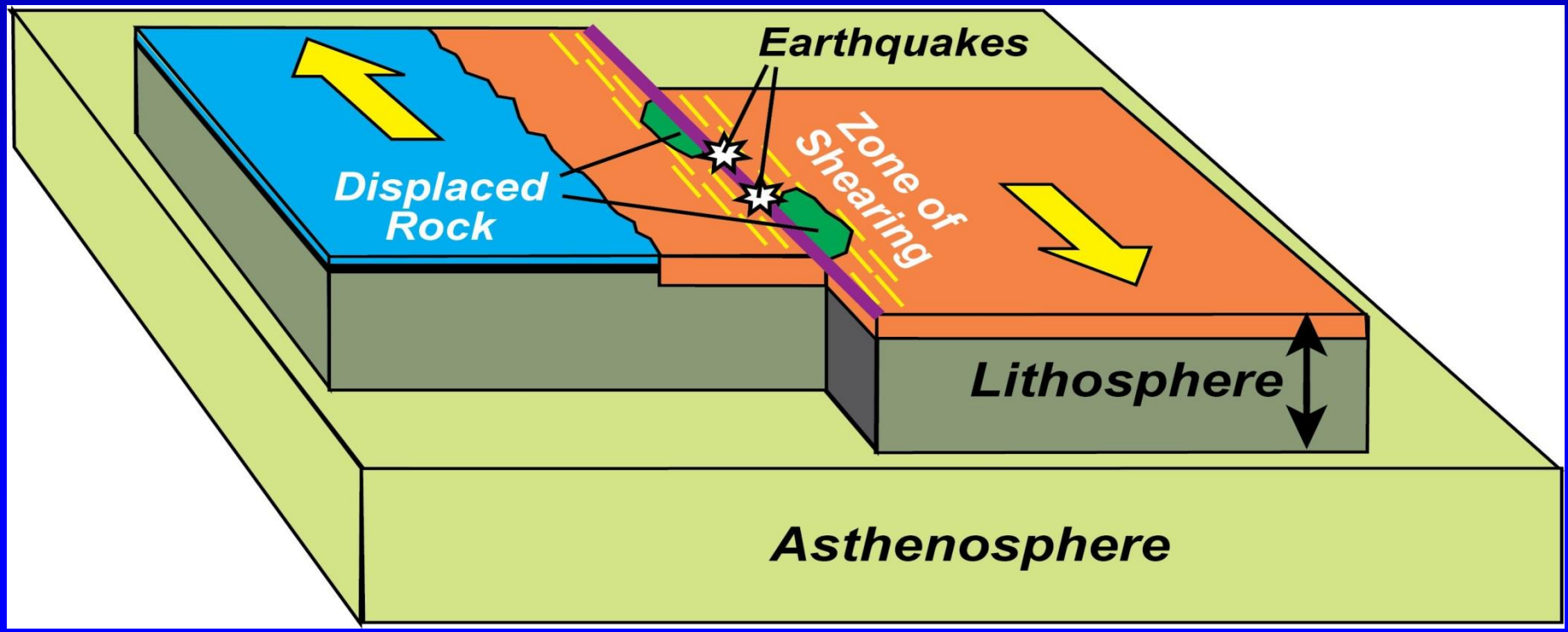
- ***Continent-Continent plate convergence (Marked by mountain belts and thrust faults)***
- تقارب لوحة القارة والقارة تتميز بالأحزمة الجبلية والفوالق



The Himalaya Mountain chain and the Tibetan Plateau are examples of this kind of margin. Both were formed when the Indian plate collided with the Eurasian plate.

Transform Plate Boundaries حدود تحولية

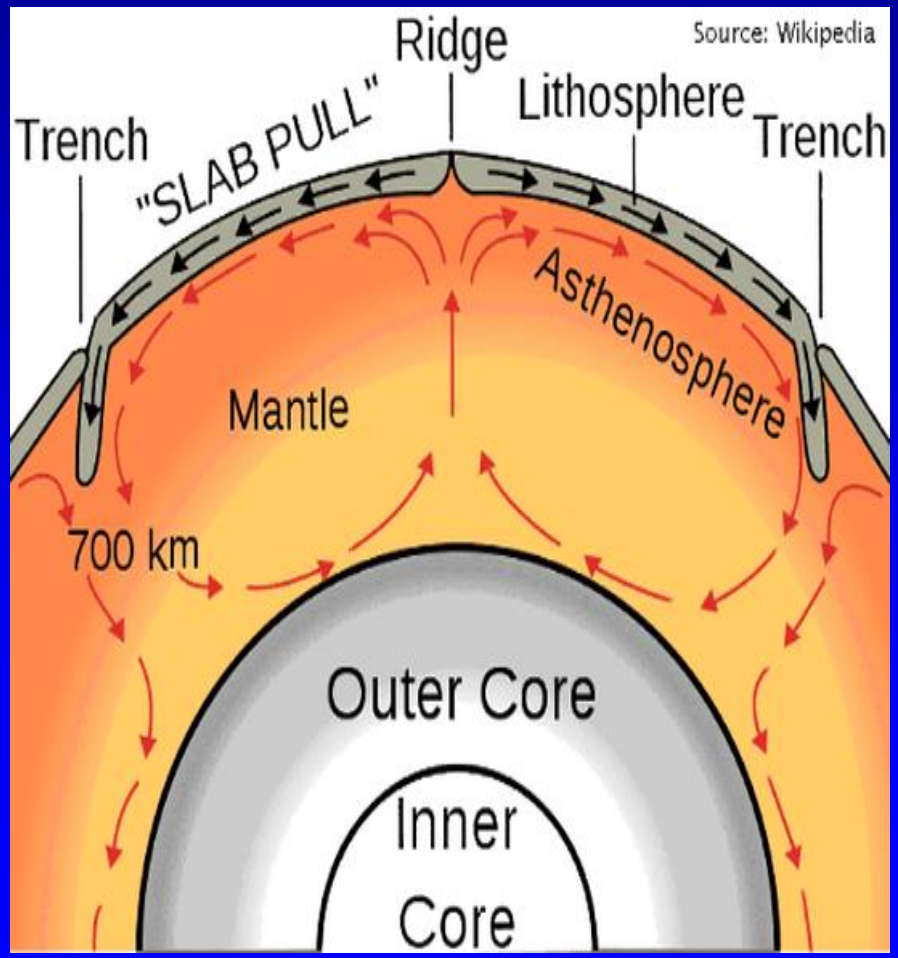
- Plates slide past one another
- Fault zones, earthquakes mark plate boundaries
- San Andreas fault in California

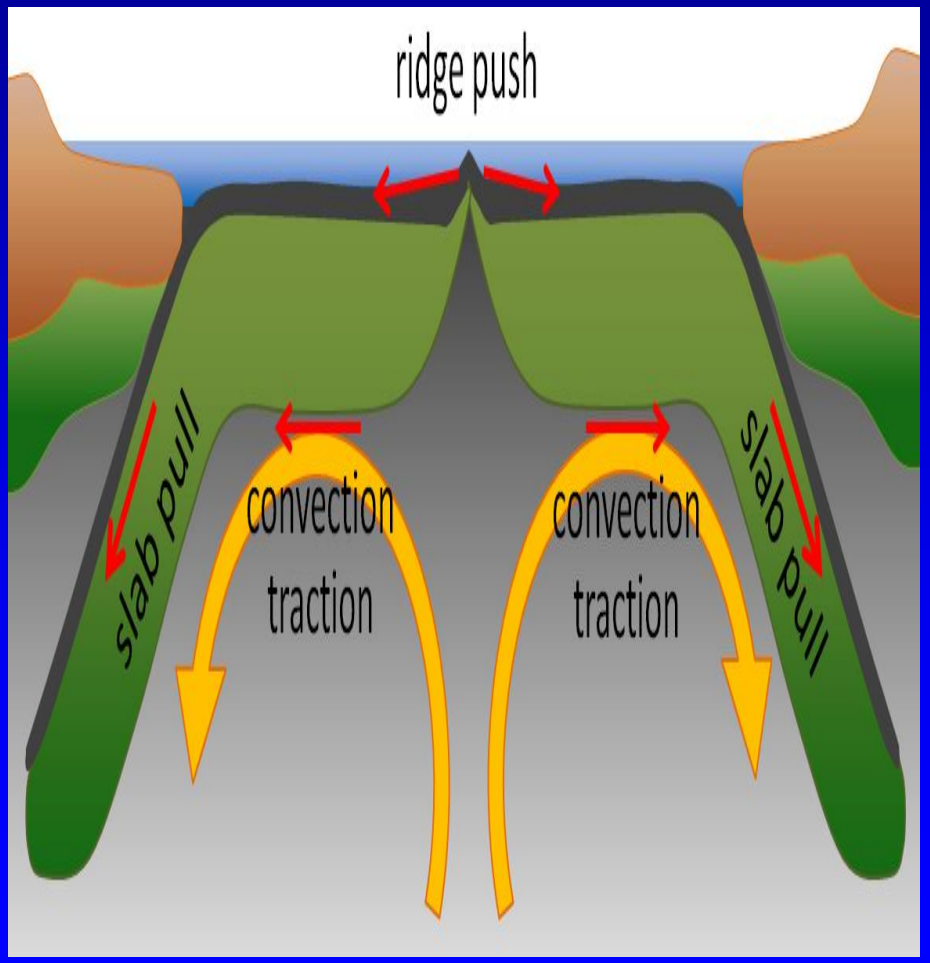


San Andreas fault 1300 km long, 30 MY between north America and pacific ocean

Causes of Plate Motions

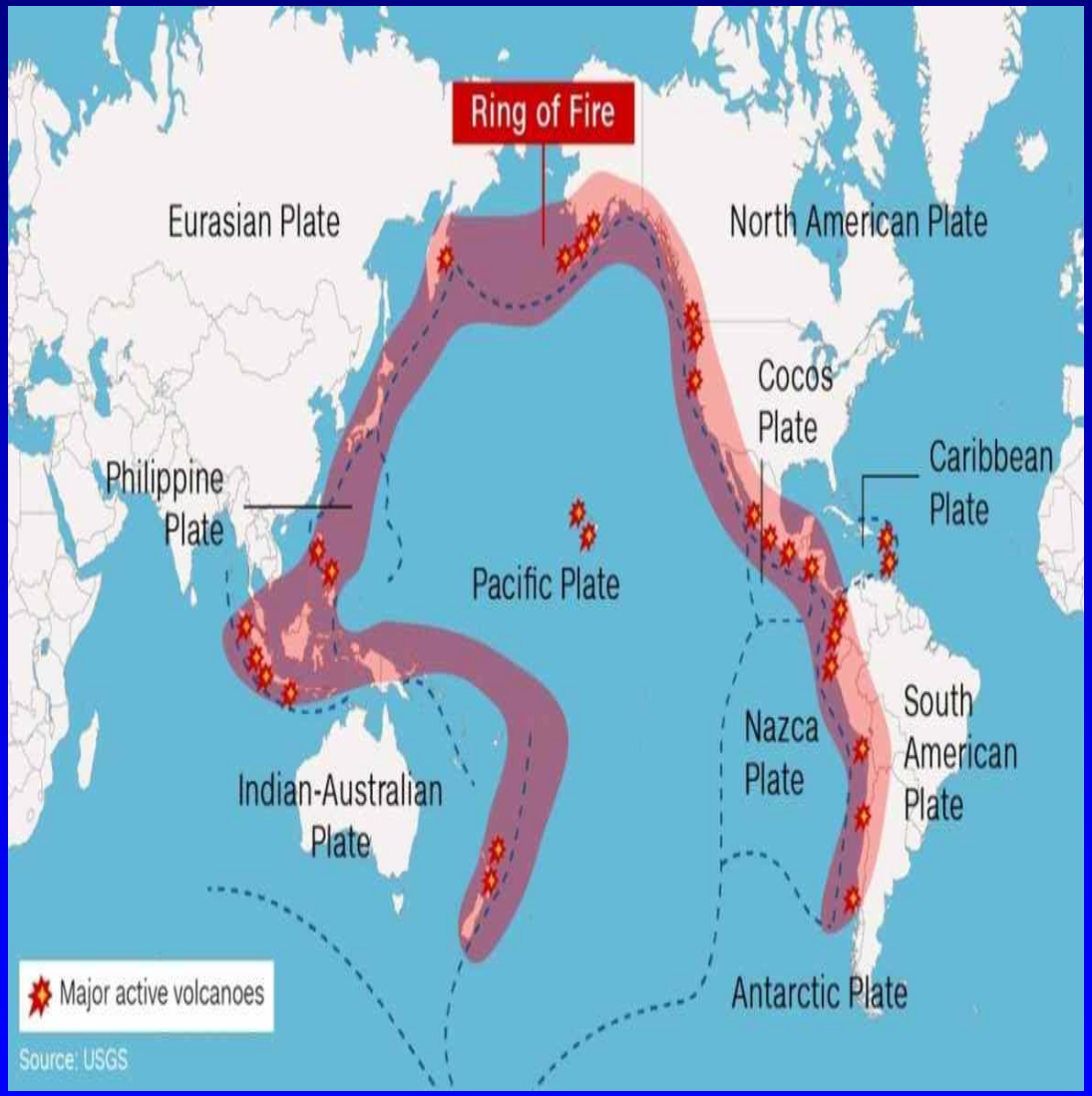
Mantle convection may be the cause **or** an effect of circulation set up by *slab-pull* and/or *ridge-push*





Pacific Ring of Fire

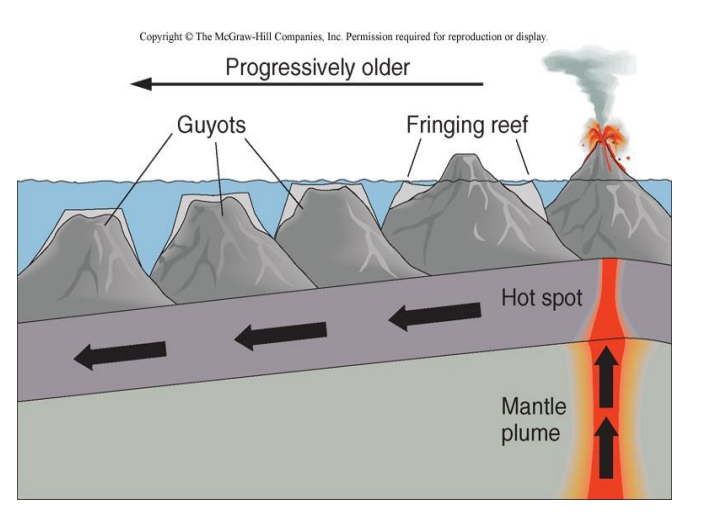
Ring حلقة of Fire, long horseshoe-shaped seismically active belt of earthquake epicenters, volcanoes, and tectonic plate boundaries that fringes the **Pacific** basin.



المحيط الهادى

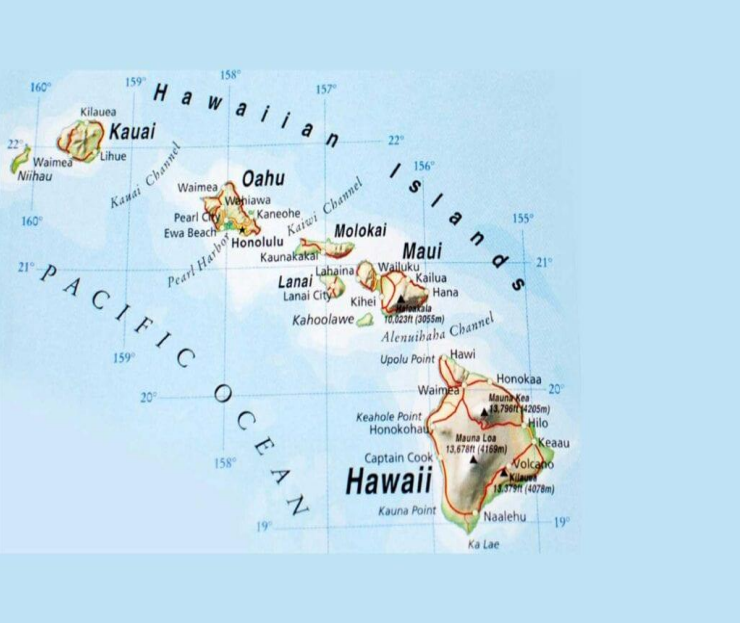
Mantle Plumes and Hot Spots اعمدة الوشاح والبقع الساخنة

Mantle plumes may form *hot spots* of active volcanism بقعة واعمدة

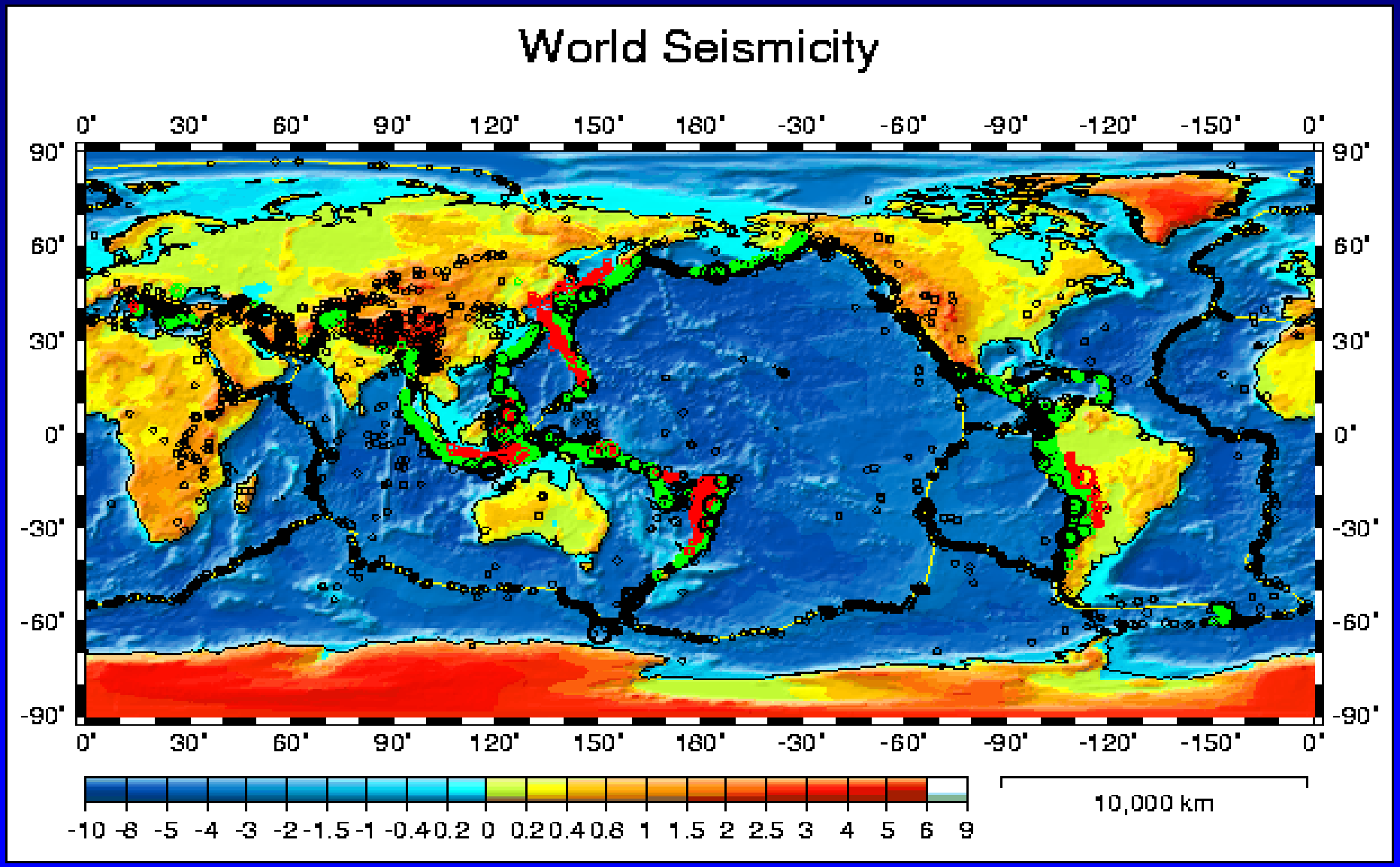


Hot spots in the interior of a plate produce *volcanic chains*

- ***Orientation* of the volcanic chain shows *direction* of plate motion over time**
- ***Age* of volcanic rocks can be used to determine *rate* of plate movement**
- **Hawaiian islands are a good example**
  - **Approximately 45 known hotspots**



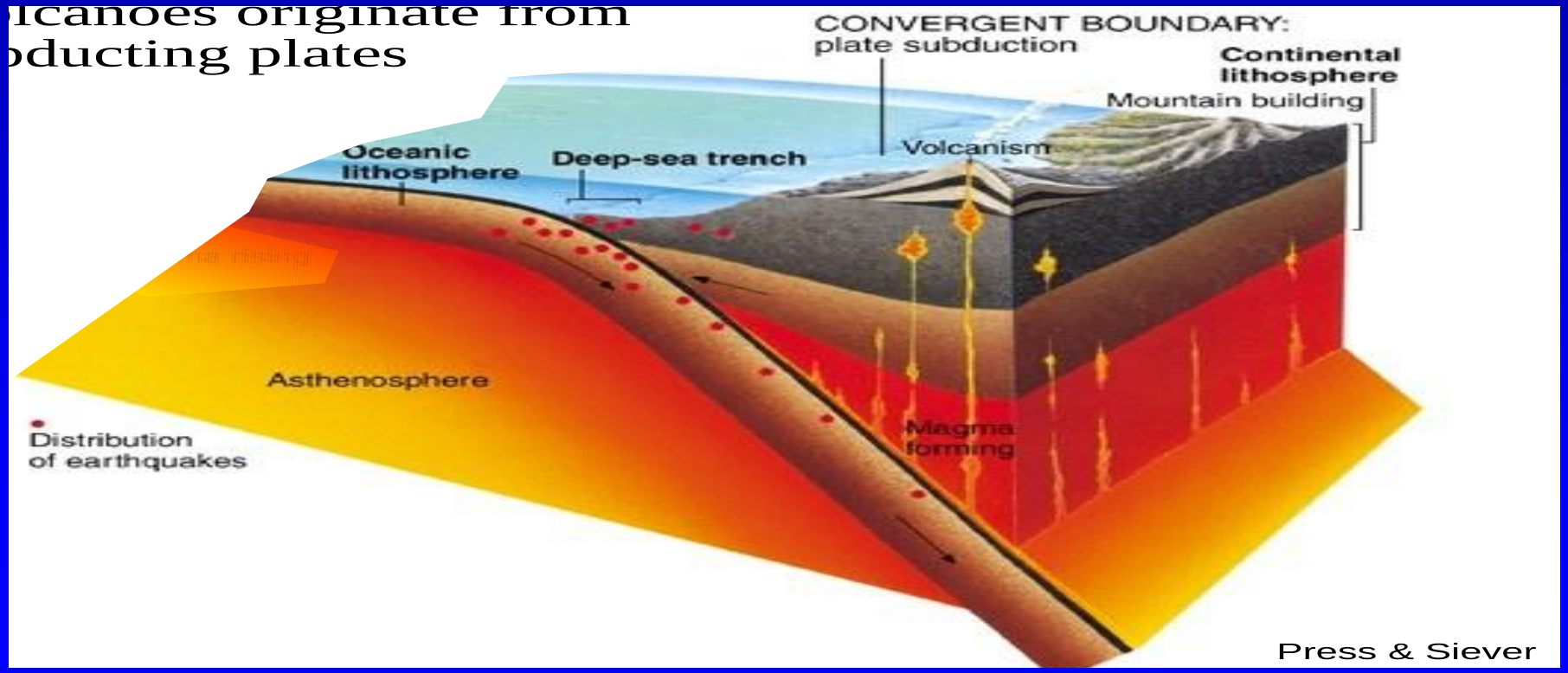
World seismicity



Consequents of Plate Movements

1-Volcanoes براكين

A volcanic eruption occurs where hot magma rises to the Earth's surface. Volcanic eruptions are common at both divergent and convergent plate boundaries



2. Earthquakes زلازل

Earthquakes are common at all three types of plate boundaries, but less common within the interior of a tectonic plate. Quakes concentrate at plate boundaries simply because those boundaries are zones of deep fractures in the lithosphere where one plate slips past another

3. Mountain Building بناء الجبال

Many of the world’s great mountain chain, formed at subduction zones. Several processes combine to build a mountain chain at a subduction zone

4-Oceanic Trenches خنادق المحيط

An oceanic trench is a long, narrow trough in the sea floor that develops where a subducting plate sinks into the mantle. Trenches are the deepest parts of the ocean basins

Some basic terminology

Island Arcs جزر قوسية: chains of volcanoes adjacent to subduction zones.

Hot spots نقاط ساخنة : Thermal plumes اعمدة (heat rising in mantle). Plates move over hot spots creating a chain of volcanoes. Hawaiian Islands, Emperor Sea Mounts

Thank You